

Sky & Space: Class 6

A year-long module for class 6 in regular school periods: from a spinning ball underfoot to India in space, built on shadow sticks, a moon diary, and one sky night.

12

SESSIONS

35

CONCEPTS TAUGHT

12 h 40 min

TOTAL

2 h

OUTDOORS

School: Middle (6–8)

AUDIENCE

Moon: start when the moon is **0–2 days old** (🌑 new → 🌒 waxing crescent) or start when the moon is **27.75–29.5 days old** (🌓 waning crescent → 🌑 new)

ASSUMED ALREADY KNOWN

Angles & the Protractor

Gravity: Things Fall

Closed under prerequisites: 2 concepts in total are taken as prior knowledge.

1 A ball, spinning

CLASSROOM · 40 MIN · DAY 3

KIT Projector · Paper, pencils, protractor

L1 Reading the Globe

The globe is Earth's honest model: a tilted ball with poles, continents and oceans. Finding India, the poles and the tilt on a globe comes before any talk of rotation or seasons.

L1 Spin & Axis

A spinning top turns around an invisible line — its axis — and the axis itself stays steady while everything else whirls. Earth is a giant top; its axis points at the pole star.

L1 Day & Night

Earth spins once in about 24 hours, so the Sun appears to rise in the east and set in the west. Day and night are a matter of which side of Earth faces the Sun.

TRAINER NOTE Darken the room, one torch, one globe, and put a sticker on your city — day and night must happen to THEM, not to a diagram. Expect the classic wrong answer that the Sun goes around us or that it 'switches off'; let a student rotate the globe and discover their sticker carrying them into shadow. Do not mention orbits, seasons or tilt-consequences yet — one revolution of the globe is the whole lesson, and it is enough.

2 The dome overhead and the Sun's road

INDOOR LAB · 1 H 20 MIN · DAY 10

KIT Paper, pencils, protractor · Gnomon / shadow stick

L1 Cardinal Directions

Finding north, south, east and west from the Sun and shadows. Sunrise and sunset directions drift through the year — due east/west only near the equinoxes.

L1 The Sky Dome

The sky looks like a dome overhead: the horizon is its rim, the zenith its top. Everything we observe is a direction on this dome, not a distance.

L1 Sun's Daily Arc

The Sun climbs from the eastern horizon, peaks toward the south (from India), and sinks west. Shadow lengths and directions trace this arc.

TRAINER NOTE Take the double period so you can step into the school ground twice, an hour apart, and mark the Sun's position against a fixed landmark — the arc has to be SEEN moving, not asserted. Fix the cardinal directions to the school building itself ('the gate is west') so every later session has a shared compass. Most children believe the Sun rises exactly east every day; plant the doubt now, the seasons session will harvest it.

3 A stick tells the time

INDOOR LAB · 40 MIN · DAY 17

KIT Gnomon / shadow stick · Measuring tape · Paper, pencils, protractor

L1 Light & Shadow

Light travels in straight lines, so an object blocks it and casts a shadow shaped like itself. Shadow direction and length depend on where the light is — the key to sundials and eclipses.

L2 Angles on the Sky

Sky positions and sizes are measured in angles: your fist at arm's length spans about 10° , the Moon half a degree. Degrees, arcminutes and arcseconds are the astronomer's ruler.

L2 Gnomon & Local Noon

A vertical stick tells time: its shadow is shortest at local noon, pointing exactly north–south. The gnomon (shanku) is astronomy's oldest instrument.

TRAINER NOTE Plant the gnomon before school and have monitors mark the shadow tip every period — by this class there is a curve on the ground and the shortest shadow points true north. Teach fist-at-arm's-length = 10° here, on the gnomon's shadow angle; the class will use it all year. The stick stays planted: it is the course's only permanent instrument, and sessions 4 and 6 come back to it.

4 How big is the Earth?

INDOOR LAB · 1 H 20 MIN · DAY 31

KIT Gnomon / shadow stick · Measuring tape · Paper, pencils, protractor · Projector

L1 Maps & Scale Models

A map shrinks the real world by a fixed scale — 1 cm standing for 1 km — keeping shapes and directions honest. Star charts, globes and solar-system walks are all the same idea.

L2 Latitude & Longitude

An address grid for the whole planet: latitude counts degrees from the equator, longitude from Greenwich. Bhopal sits near 23°N, 77°E — and your latitude is written in your sky.

L2 Earth is a Sphere

Ships sinking hull-first, new stars rising as you travel south, and Earth's always-round shadow on the eclipsed Moon — classical proofs, no spacecraft needed.

L3 Measuring the Earth

Eratosthenes sized the Earth around 240 BCE with two sticks and a noon shadow angle. Two schools at different latitudes can repeat it today over WhatsApp.

TRAINER NOTE Start from the school map they drew in geography and shrink outwards — city, country, globe — so scale is a habit before it is a number. Then Eratosthenes: same stick, same noon, a partner school a few hundred km north or south, and the class computes the circumference of the planet themselves. Arrange the partner school weeks in advance; a WhatsApp photo of their noon shadow is all you need. If the numbers come out within 15% the room goes silent — do not fill the silence.

5 Why summer? (Not what you think)

CLASSROOM · 40 MIN · DAY 45

KIT Projector · Paper, pencils, protractor

L2 Seasons

Earth's axis is tilted 23.4°, so through the year each hemisphere leans toward or away from the Sun — changing day length and the Sun's height. Distance from the Sun is not the cause.

L2 Equinoxes & Solstices

Four turning points of the year: two equinoxes when day equals night, two solstices when the Sun's arc is highest or lowest. They anchor seasons and calendars worldwide.

TRAINER NOTE Before teaching anything, have every child write one sentence explaining summer. Most will say the Earth gets closer to the Sun — keep the slips. Globe and lamp, tilt fixed, walk the orbit: let a student notice that 'closer' would make BOTH hemispheres hot at once, while Australia's December beaches say otherwise. Return the slips at the end and let them mark their own. The gnomon's noon shadow, now weeks long in your logbook, shows the solstice drift for free.

6

The noon shadow project

SOLAR SESSION · 40 MIN · DAY 52

SKY

needs **Any moon** · The Sun

KIT

Gnomon / shadow stick · Measuring tape · Paper, pencils, protractor

L2

Tropics & Polar Circles

The Tropic of Cancer — which crosses India through MP and Gujarat — marks where the noon Sun can stand exactly overhead. These lines are the axial tilt drawn onto the map.

L2

Zero Shadow Day

Twice a year, everywhere between the tropics, the noon Sun stands exactly overhead and shadows vanish for a moment. India's favourite mass astronomy experiment — date depends on your latitude.

TRAINER NOTE A daytime outing to the faithful stick: measure the noon shadow carefully and compute your latitude's distance from the Tropic of Cancer. If your school sits south of the Tropic, find your actual zero-shadow date from the tables and move THIS session to match it — a vertical stick with no shadow is the single most photographed moment of the year. North of the Tropic, teach it as the shadow that never quite dies, and why that line on the map is drawn where it is. Never let anyone stare at the Sun; the shadow is the instrument.

7

Start your moon diary

CLASSROOM · 40 MIN · DAY 59

KIT

Paper, pencils, protractor

L1

Observing the Moon

The Moon is visible in the day as often as at night, rises about 50 minutes later each day, and shows changing shapes. Just watching it for a month is a complete experiment.

TRAINER NOTE One concept, one period, and a month of homework: every child sketches the Moon each clear night — shape, time, and which side is lit — in a printed 30-box diary. Teach them that the Moon is up in the DAYTIME half the month, which most adults do not know, so morning sketches count. This session exists to start the clock: the phases lesson is deliberately four weeks away, because the Moon is the only teacher of its own rhythm. Chase the diaries weekly or the lab on day 87 falls flat.

8 Phases, from inside the orbit

INDOOR LAB · 1 H 20 MIN · DAY 87

KIT Paper, pencils, protractor

L2 Models in Science

A globe, a map, a diagram of the Sun — none are the real thing; all are models that answer some questions well and others not at all. Science improves by testing models against observation.

L2 Phases of the Moon

The Moon is always half-lit by the Sun; the phase is how much of that lit half faces us, cycling in about 29.5 days. Phase, rise time and Sun–Moon angle all lock together.

L2 Moon Orbits Earth

The Moon circles Earth roughly monthly at about 384,000 km — around 30 Earth-diameters away. Its orbital motion is what drives the phase cycle and its nightly 13° eastward drift.

TRAINER NOTE Darken the hall: one lamp is the Sun, every child holds a ball at arm's length and turns on the spot — each of them is the Earth watching their own moon wax and crescent. Phases must be seen from inside the orbit; a diagram drawn from above teaches the teacher, not the child. Then lay their month of diaries end to end and match diary to ball. The misconception to kill on sight: 'phases are the Earth's shadow'. Do not kill it with words — ask where the shadow would have to be, and let the lamp answer. Close by naming what you just did: the lamp is not the Sun and the ball is not the Moon, yet they answered a real question — that is what a scientific model IS, and this class will use models all year.

9 The sky night

OBSERVING NIGHT · 1 H 20 MIN · DAY 94

SKY needs **Thin crescent** · The Moon, Constellations, Bright stars, Horizon / cardinal points

KIT Star chart / planisphere · Red torch

L1 Stars Differ

Stars are not identical dots: they differ in brightness and in colour — steely blue Vega, golden Arcturus, red Betelgeuse. These differences are real physics, visible to the naked eye.

L1 Constellations

Star patterns used as sky landmarks — 88 official constellations today, with older patterns in every culture. They are direction labels, not physical groups.

L2 Diurnal Motion

The whole starry sky wheels westward through the night, circling a fixed point — the celestial pole. This is Earth's rotation seen from the inside.

L2 The Pole Star

Polaris (Dhruva) sits nearly at the north celestial pole, so it barely moves and always marks north. Its altitude above the horizon equals your latitude.

L2 Indian Star & Constellation Names

Dhruva (Polaris), Saptarshi (the Big Dipper), Arundhati–Vasishtha (the Mizar–Alcor pair), Rohini (Aldebaran), Chitra (Spica): the Indian sky map atop the same stars.

TRAINER NOTE The one evening the school will give you — spend it on the naked eye, not on a telescope queue. A crescent moon greets them early (point out the earthshine even though you are not formally teaching it) and sets before it can wash out the constellations. Anchor two constellations, then return to them after an hour so the class SEES the sky wheel; only then reveal the one star that did not move. Expect disbelief that Dhruva is faint — that surprise is the lesson. Give the Indian names alongside the Greek ones as equals, not as trivia. Have a cloud date agreed with the school in advance.

10 When shadows swallow the Moon and Sun

INDOOR LAB · 1 H 20 MIN · DAY 101

KIT Projector · Paper, pencils, protractor

L1 How We See

We see things only when light from them enters our eyes — the eye receives light, it doesn't send anything out. No light, no seeing; more light gathered, fainter things seen.

L2 Lunar Eclipses

At full moon, the Moon can pass through Earth's shadow and turn copper-red — sunset light bent through our atmosphere. Safe to watch, visible from the whole night side.

L2 Solar Eclipses

At new moon, the Moon's small shadow can sweep across Earth, briefly hiding the Sun. Totality happens because the Moon and Sun span the same half-degree — a cosmic coincidence.

TRAINER NOTE Same lamp and balls as the phases lab, deliberately — an eclipse is what happens on the rare month the three actually line up, and the class already owns the model that shows why it is rare. Contrast against phases explicitly: last month's wrong answer ('Earth's shadow') is this month's right one, and saying so out loud inoculates both. Teach eye safety with full weight here: a solar eclipse is the one astronomical event that can injure a child, and the pinhole projector they build in the last twenty minutes is their licence to watch one. Address the eclipse-food superstitions directly and without mockery — the children will hear them at home.

11 The wanderers and the family

INDOOR LAB · 1 H 20 MIN · DAY 122

KIT Projector · Paper, pencils, protractor · Measuring tape

L2 Big Numbers & Powers of Ten

A lakh, a crore, a billion, 10^8 — big numbers need both names and notation. Astronomy runs on powers of ten; without them, its distances are just meaningless words.

L2 Planets: the Wanderers

A few bright 'stars' drift against the fixed constellations from week to week — the planets. They shine steadily while true stars twinkle.

L2 The Solar System Family

Eight planets, their moons, dwarf planets, asteroid belt and Kuiper belt — who's who, their order, and their wildly different sizes and characters.

L3 Scale of the Solar System

With Earth as a peppercorn, the Sun is a football 25 m away and Neptune is 750 m down the road. The emptiness is the real lesson; the AU is the ruler.

TRAINER NOTE Start from the sky night: everything they saw stayed fixed in its constellation — except, had they watched for weeks, a few bright 'stars' that drift. That drift, not a poster of eight balls, is what a planet IS, and it is why five of them have been known since before history. Then take the family tour, but keep insisting on the difference between what we see (dots that wander) and what we know (worlds). Walk the planet order down the corridor first — sequence only. Then do it honestly: Earth a peppercorn, the Sun a football 25 m away, Neptune 750 m down the road. Every textbook poster lies about this, and the walk is the one lesson about the solar system nobody forgets: it is made of emptiness.

12 India in space

REVIEW & ASSESS · 1 H 20 MIN · DAY 150

KIT Projector · Paper, pencils, protractor

L2 India in Space

Chandrayaan-3 landed at the lunar south pole, Mangalyaan orbited Mars on a first attempt, Aditya-L1 stares at the Sun, AstroSat scans the X-ray sky — and Gaganyaan comes next.

L3 Aryabhata's Astronomy

In 499 CE Aryabhata taught that Earth rotates, gave π to four decimals, built sine tables, and explained eclipses by shadows — not demons. The namesake of this foundation.

TRAINER NOTE The closing session joins the two ends of the year: Aryabhata computed the Earth's spin and explained eclipses by shadows fifteen centuries before the satellite named after him flew — and the class has personally redone both of those pieces of work with a stick and a lamp. Run it as review-by-storytelling: Chandrayaan needs their moon diary, Aditya needs their eye-safety rules, Mangalyaan needs their wanderers. End by having each child present one measurement THEY made this year. The message to land: this subject is not finished, and it is theirs.

FULL KIT LIST

Gnomon / shadow stick (<i>consumable</i>)	sessions 2, 3, 4, 6
Measuring tape	sessions 3, 4, 6, 11
Projector	sessions 1, 4, 5, 10, 11, 12
Red torch	sessions 9
Star chart / planisphere	sessions 9
Paper, pencils, protractor (<i>consumable</i>)	sessions 1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12